

IH-53 EV wiring list

Physical wires taken from [IH53EV.kicad_sch](#). Size/type is the schematic text on that run when present. **Color prefers the component wiring-harness pigtail** and is carried through splices; schematic stroke color is ignored when a harness color exists.

Inline disconnects (Anderson, DJ70xx, Weather Pack, Deutsch, USB) are drawn as two-sided symbols. Each side is a separate cable termination; mating the housings is not a wire.

Orion BMS 2 Main I/O on this vehicle is the **CWHMIO 18 AWG** harness (Sumitomo 8240-4892 crimps), not 22 AWG.

Color rules

1. If a factory harness / pigtail has a color, that color is the color of the whole run, including splices and bulkhead connectors.
2. When two pigtails meet (for example TSM2500 CAN into Orion CAN1), **keep the Orion (or other "bus owner") color on the trunk**; the other pigtail keeps its factory color only up to the splice.
3. Traction **2/0 DLO** is **orange** (EV HV jacket). Polarity is **red** (+) / **black** (–) adhesive heatshrink at each lug — not a red/black jacket.
4. 12 V SGX / TXL battery feeds have no colored pigtail: **red** = +, **black** = –.
5. Vehicle **DCIS pack HV** (10 AWG / 18 AWG HV from the 3-way toward the pack) is **red** = +, **black** = –. Factory DCIS pigtail stays **yellow / gray** only up to [dj70310-6.3-1](#).
6. Orion CWHMIO colors: pin 1 (Always_On) = **green**, pin 2 (READY) = solid **blue**, and pin 3 (CHARGE) = solid **red** (2 and 3 from [Orion's CANdapter troubleshooting page](#); pin 1 from this harness). Pins 6, 7, 13, 14 from this vehicle's harness. Remaining pins still **verify on harness card**.

Wire key (from schematic)

Code	Meaning
AWG	conductor size
TXL	12 V / 48 V primary (GXL OK, same AWG)
SGX	12 V battery cable (10 AWG)
HV	600–1000 V DC pack (silicone / EV)
AC	J1772 / T92 mains (600 V)
300VAC	RAC02 tap only (schematic may say 18AWG AC — same run)
DLO	traction / pack 2/0

Harness color map (factory pigtails)

Device	Pin / lead	Color	Size	Source
Orion CWHMIO	1 Always_On_Power	green	18 AWG	this harness
Orion CWHMIO	2 READY_Power	blue	18 AWG	Orion troubleshooting
Orion CWHMIO	3 CHARGE_Power	red	18 AWG	same
Orion CWHMIO	6 Charger_Safety	red/white	18 AWG	this harness
Orion CWHMIO	7 DISCHARGE_enable	blue/white	18 AWG	this harness
Orion CWHMIO	12 Ground	black (typical; verify)	18 AWG	CWHMIO; ground convention
Orion CWHMIO	13 J1772_Control_Pilot	orange/white	18 AWG	this harness

Device	Pin / lead	Color	Size	Source
Orion CWHMIO	14 J1772_Proximity_Detect	brown/white	18 AWG	this harness
Orion CWHMIO	17 CAN1_Shield	drain / bare	shield	CWHMIO CAN1 STP
Orion CWHMIO	18 CAN1_H	CAN1 pair (gray jacket STP; inner <i>verify</i>)	18 AWG pair, 12 ft	purchasing guide
Orion CWHMIO	19 CAN1_L	CAN1 pair (with pin 18)	18 AWG pair, 12 ft	same
TSM2500 AC	L / N / PE	brown / blue / green-yellow	2.5 mm ² (~14 AWG)	TSM2500 manual
TSM2500 DC	+ / –	red / black	6 mm ² (~10 AWG HV , not TXL)	same (SB50); pack voltage
TSM2500 CAN	H / L	green/white / blue/white	0.5 mm ²	same; Orion TSM2500 note (splice onto Orion CAN1 colors)
DCIS output	+ / –	red / black	6.0 mm ² (~10 AWG)	DCIS schematic
DCIS input	+ / –	factory yellow / gray ; vehicle red / black	10 AWG HV (factory pigtail 1.5 mm ²)	factory pigtail to 3-way; vehicle HV is red/black
DCIS key	Key_Switch_Control	green	0.5 mm ²	same
HyPer 9 encoder (not on this sheet)	K1-35 / 21 / 33 / 9	red / yellow / green / black	18 / 20 AWG	HyPer manual
HyPer 9 K1 CAN, K1-24, K1-25, K1-26	—	<i>verify on K1 print</i> (colors vary by kit)	18–20 AWG	HyPer manual: identify by pin number
HyPer 9 B+ / B–	DC lugs	orange jacket; red / black HS	2/0 DLO	EV orange; polarity by heatshrink
CANdapter USB	cable	black jacket (USB 2.0)	USB	CANdapter

Connector reference

Device	Mating connector	Contacts / wire	Source
Orion BMS 2 Main I/O	TE 1376360-1	18 AWG + Sumitomo 8240-4892 (this harness)	user; Orion wiring manual
Orion voltage taps (cell taps, not drawn as wires on this sheet)	TE 1318389-1	22 AWG + TE 1123343-2 (gold); group colors orange / red / yellow, grounds black	same
HyPer 9 X1 K1	TE AMPSEAL 35-pin 776164-1, pins 770854-1	typically 18–20 AWG	HyPer System User Manual
HyPer 9 B+ / B–	inverter DC lugs	2/0 DLO, ring terminal	schematic
TSM2500 AC input	DJ7031-4.8 (DJ7031-4.8 ; 4.8-11 / 4.8-21)	2.5 mm ² (~14 AWG) brown / blue / green-yellow	TSM2500 manual; schematic

Device	Mating connector	Contacts / wire	Source
TSM2500 DC output	Anderson SB50 (AndersonSB50)	6 mm ² (~10 AWG HV) red / black	same
TSM2500 CAN	charger CAN pair	0.5 mm ² green/white + blue/white	same
DCIS 600 W output	DJ7021-8-21 (DJ7021-8-1)	6.0 mm ² (~10 AWG TXL) red / black	DCIS schematic; schematic
DCIS 600 W input / enable	dj70310-6.3-11 (dj70310-6.3-1)	10 AWG HV vehicle red / black (factory 1.5 mm ² yellow / gray); 18 AWG HV / 0.5 mm ² green key	this vehicle / DCIS schematic
CANdapter CAN	DE-9 (DB9)	twisted pair to pins 7 (CAN_H) and 2 (CAN_L)	CANdapter manual
CANdapter USB	USB-B on adapter	USB cable	CANdapter manual
Blue Sea 5032 buses	#10-32 stud	ring terminal	Blue Sea 5032
Blue Sea 5032 circuits	#8-32 screw	ring or spade	same
J1772 inlet	SAE J1772 socket	AC pins + CP / PP	schematic
Anderson 350 A / 600 V	SB350 housing	2/0 contacts	schematic
Anderson SB50	SB50 housing	6–10 AWG contacts	TSM2500 DC only
DJ7021-8-21	2-way, 8.0 mm	DCIS 12 V output	schematic (included with DCIS)
dj70310-6.3-11	3-way, 6.3 mm	DCIS pack input / key	schematic (included with DCIS)
DJ7031-4.8	3-way, 4.8 mm	TSM2500 AC	schematic (included with TSM2500)
Delphi Weather Pack 4	Weather Pack 4-way	18 AWG (Orion colors carried through)	schematic
Deutsch DTM 3	DTM 3-way	16–22 AWG	schematic
BB CANOP isolator	terminal-block CAN / power	no factory pigtail color	Advantech BB-CANOP
HV terminal blocks	5/16" and 1/4" studs	2/0 lugs on 5/16"; smaller HV on 1/4"	schematic
Gigavac GV200QA-1	M8 HV terminals; coil X1/X2	2/0 lugs; coil from K1 harness	schematic
T92P11D22-12	0.250" QC	14 AWG AC; 18 AWG coil from Orion	TE T92
T9AP1D52-12	0.250" QC	18 AWG where tied to Orion	TE T9A
G9EJ-1-E-UVD DC12	0.250" QC	18 AWG HV contacts (polarized 3+/4-); 18 AWG TXL coil	Omron G9EJ
Cole Hersee 24213	stud contacts; QC coil	10 AWG SGX contacts	schematic
MAXI 5006	MAXI fuse holder lugs	10 AWG SGX / 10 AWG TXL	schematic
A25X500-4	bolted fuse	2/0 DLO lugs	schematic

1. Traction pack (2/0 DLO)

Tesla modules **BT2–BT6** in series. Inter-module runs are unlabeled next to the modules; 2/0 DLO labels sit on that pack string. Jacket is **orange** (EV HV). Mark every lug with **red** (+) or **black** (–) adhesive heatshrink.

ID	Size / type	Color	End A	A termination	End B	B termination
HV-01	2/0 DLO	orange, black HS	BT2 –	module – lug	TerminalBlock2 5/16"	5/16" stud, ring lug
HV-02	2/0 DLO	orange, red HS	BT2 +	module + lug	BT3 –	module – lug
HV-03	2/0 DLO	orange, red HS	BT3 +	module + lug	BT4 –	module – lug
HV-04	2/0 DLO	orange, red HS	BT4 +	module + lug	BT5 –	module – lug
HV-05	2/0 DLO	orange, red HS	BT5 +	module + lug	BT6 –	module – lug
HV-06	2/0 DLO	orange, red HS	BT6 +	module + lug	F4 (A25X500-4)	bolted fuse pad
HV-07	2/0 DLO	orange, red HS	F4	bolted fuse pad	SW1 (Kill Switch HVBD6AXR)	switch HV terminal
HV-08	2/0 DLO	orange, red HS	SW1	switch HV terminal	TerminalBlock1 5/16"	5/16" stud, ring lug
HV-09	2/0 DLO	orange, red HS	TerminalBlock1 5/16"	5/16" stud, ring lug	Anderson350A600V1 right +	SB350 contact (2/0)
HV-10	2/0 DLO	orange, black HS	TerminalBlock2 5/16"	5/16" stud, ring lug	Anderson350A600V1 right –	SB350 contact (2/0)
HV-11	2/0 DLO	orange, red HS	Anderson350A600V1 left +	SB350 contact (2/0)	TerminalBlock3 5/16" (pack + bus)	5/16" stud, ring lug
HV-12	2/0 DLO	orange, red HS	Anderson350A600V1 left –	SB350 contact (2/0)	HyPer 9 B+	inverter B+ lug
HV-13	2/0 DLO	orange, black HS	HyPer 9 B–	inverter B– lug	G1 (GV200QA-1) A2–	Gigavac M8 HV terminal
HV-14	2/0 DLO	orange, red HS	G1 A1+	Gigavac M8 HV terminal	TerminalBlock3 5/16" (pack + bus)	5/16" stud, ring lug

SB350 left housing mates to right housing: pack + bus ↔ TB1, inverter B+ ↔ TB2–.

Cell taps on BT2–BT6 are **not** wired on this sheet. Orion tap harness is 22 AWG to TE 1318389-1 (orange / red / yellow groups, black grounds). Schematic notes **internally termination** at the modules.

2. Charger HV DC

TSM2500 DC is the included **Anderson SB50** (**AndersonSB50**). Mating is not a wire: left housing is the charger pigtail, right housing is the vehicle. Pigtail colors **red / black** carry through. Vehicle runs are **10 AWG HV** (labeled). Factory pigtail is 6 mm² (~10 AWG) unlabeled on the charger side of the SB50.

SB50 mates top-to-top: charger + (left top) → vehicle right top → 32 A 2-pole → TerminalBlock1 1/4" (pack +). Charger – (left bottom) → vehicle right bottom → Circuit_Breaker3 → TerminalBlock2 1/4" (pack –).

ID	Size / type	Color	End A	A termination	End B	B termination	Source
CH-DC+	6 mm ² charger pigtail	red	TSM2500 DC_Output_+	charger DC +	AndersonSB50 left top	SB50 contact	TSM2500 pigtail
CH-DC-	6 mm ² charger pigtail	black	TSM2500 DC_Ouput_-	charger DC -	AndersonSB50 left bottom	SB50 contact	TSM2500 pigtail
CH-DC+V	10 AWG HV	red	AndersonSB50 right top	SB50 contact	2Pole_1000V_DC_32Amp1 B	breaker HV terminal	carry TSM2500 red
CH-DC+2	10 AWG HV	red	2Pole_1000V_DC_32Amp1 A	breaker HV terminal	TerminalBlock1 1/4"	1/4" stud, ring lug	carry TSM2500 red
CH-DC-V	10 AWG HV	black	AndersonSB50 right bottom	SB50 contact	Circuit_Breaker3 B	breaker HV terminal	carry TSM2500 black
CH-DC-2	10 AWG HV	black	Circuit_Breaker3 A	breaker HV terminal	TerminalBlock2 1/4"	1/4" stud, ring lug	carry TSM2500 black

3. DC-DC converter (pack → 12 V)

DCIS factory connectors are drawn: **dj70310-6.3-11** (**dj70310-6.3-1**) 3-way input/key, **DJ7021-8-21** (**DJ7021-8-1**) 2-way 12 V output. Mating is not a wire. Factory pigtail stays **yellow / gray / green** only to the 3-way. Vehicle pack HV from that connector is **red / black**. Key stays **green**. 12 V output is **red / black**.

Pack input vehicle runs are **10 AWG HV** (same cable as charger DC). Factory pigtail to the 3-way is 1.5 mm². Key is pack voltage (DCIS: "key switch control voltage is consistent with the input voltage"), so the vehicle key run and the K4 tap are **18 AWG HV**, not TXL. 12 V output factory side is **10 AWG TXL** (labeled). If 10 AWG will not enter the 6.3 mm contacts, splice a short 14–16 AWG tail into the housing.

dj70310 pin order top → bottom: Input+ / Input- / Key. DJ7021 pin order top → bottom: Output+ / Output-. Left = DCIS pigtail, right = vehicle on the 3-way; **right = DCIS pigtail, left = vehicle** on the 2-way (as drawn).

ID	Size / type	Color	End A	A termination	End B	B termination	Source
DC-IN0+	1.5 mm ² factory pigtail	yellow	DCIS Input+	DCIS 3-pin	dj70310-6.3-1 left top	6.3 mm contact	DCIS yellow
DC-IN0-	1.5 mm ² factory pigtail	gray	DCIS Input-	DCIS 3-pin	dj70310-6.3-1 left mid	6.3 mm contact	DCIS gray
DC-KEY0	0.5 mm ² factory pigtail	green	DCIS Key_Switch_Control	DCIS 3-pin	dj70310-6.3-1 left bottom	6.3 mm contact	DCIS green
DC-IN+	10 AWG	red	dj70310-6.3-1 right top	6.3 mm contact	Circuit_Breaker4 A	breaker HV terminal	vehicle HV+

ID	Size / type	Color	End A	A termination	End B	B termination	Source
	HV						
DC-K4TAP	18 AWG HV	red	splice on DC-IN+	—	K4 pin 3 (+)	G9EJ #250 QC	pack + to key relay
DC-IN+2	10 AWG HV	red	Circuit_Breaker4 B	breaker HV terminal	TerminalBlock1 1/4"	1/4" stud	vehicle HV+
DC-IN-	10 AWG HV	black	dj70310-6.3-1 right mid	6.3 mm contact	2Pole_1000V_DC_10Amp1 A	breaker HV terminal	vehicle HV-
DC-IN-2	10 AWG HV	black	2Pole_1000V_DC_10Amp1 B	breaker HV terminal	TerminalBlock2 1/4"	1/4" stud	vehicle HV-
DC-KEY	18 AWG HV	green	dj70310-6.3-1 right bottom	6.3 mm contact	K4 pin 4 (-)	G9EJ #250 QC	carry DCIS green
DC-OUT+	10 AWG TXL	red	DCIS Output+	DCIS 2-pin	DJ7021-8-1 right top	8.0 mm contact	DCIS red 6.0 mm ²
DC-OUT-	10 AWG TXL	black	DCIS Output-	DCIS 2-pin	DJ7021-8-1 right bottom	8.0 mm contact	DCIS black 6.0 mm ²

Precharge path from pack + into the inverter (no pigtail). Schematic unlabeled; **18 AWG HV** (T9A QC), **red**. Circuit_Breaker2 is **1-pole 1000 V DC 10 A**.

ID	Size / type	Color	End A	A termination	End B	B termination
PC-01	18 AWG HV (unlabeled)	red	TerminalBlock3 1/4"	1/4" stud	Circuit_Breaker2	breaker HV terminal
PC-02	18 AWG HV (unlabeled)	red	Circuit_Breaker2	breaker HV terminal	K2 (T9AP1D52-12) pin 4	T9A QC
PC-03	18 AWG HV (unlabeled)	red	K2 pin 3	T9A QC	HyPer 9 Precharge/Key_Switch_In (K1-24)	AMPSEAL 776164-1 pin 24

K1-24 is the only HyPer 9 K1 pin that sees pack voltage on a non-isolated X1 (HyPer manual). If the K1 pigtail for pin 24 is a different color, use that K1 color instead of red from the splice to the AMPSEAL.

4. J1772 / charger AC

TSM2500 AC is the included **DJ7031-4.8** (**DJ7031-4.8**). Right housing is the charger pigtail (**brown / blue / green-yellow**); left is vehicle **14 AWG AC**. Those colors carry through the T92 to the J1772 inlet.

RAC02-12SE/277 is 2 W (≈ 47 mA at 115 VAC). Do **not** run 14 AWG AC to F7 / RAC02. Tap **18 AWG 300VAC** off the L1/L2 mains and fuse it at F7 (F5 $\times$ 20, 1 A / Recom T1A slow-blow).

DJ7031 pin order top $\rightarrow$ bottom: L / N / PE.

ID	Size / type	Color	End A	A termination	End B	B termination
AC-L1	14 AWG AC	brown	J1772 L1	J1772 inlet L1	T92 pin 24	0.250" QC
AC-L1-F7	18 AWG 300VAC	brown	splice on AC-L1	—	F7 (F5×20, 1 A max)	fuse clip
AC-L1-RAC	18 AWG 300VAC	brown	F7	fuse clip	RAC02 VAC_IN(L)	RAC02 AC pin
AC-L2	14 AWG AC	blue	J1772 L2/N	J1772 inlet L2/N	T92 pin 14	0.250" QC
AC-L2-TAP	18 AWG 300VAC	blue	splice on AC-L2	—	RAC02 VAC_IN(N)	RAC02 AC pin
AC-GND	14 AWG AC	green-yellow	J1772 Equipment Ground	J1772 inlet PE	DJ7031-4.8 left bottom <i>and</i> chassis GND	4.8 mm contact / chassis
AC-CHG-L	14 AWG AC	brown	T92 pin 23	0.250" QC	DJ7031-4.8 left top	4.8 mm contact
AC-CHG-N	14 AWG AC	blue	T92 pin 13	0.250" QC	DJ7031-4.8 left mid	4.8 mm contact
AC-CHG-L0	2.5 mm ² charger pigtail	brown	DJ7031-4.8 right top	4.8 mm contact	TSM2500 AC_Input_Line	charger AC pin 1
AC-CHG-N0	2.5 mm ² charger pigtail	blue	DJ7031-4.8 right mid	4.8 mm contact	TSM2500 AC_Input_Neutral	charger AC pin 2
AC-CHG-PE0	2.5 mm ² charger pigtail	green-yellow	DJ7031-4.8 right bottom	4.8 mm contact	TSM2500 AC_Input_Earth_Ground	charger AC pin 3

T92P11D22-12 (**Power_Relay1**) closes L1/L2 to the charger when the BMS charge coil is on.

5. 12 V vehicle battery and Blue Sea 5032

Battery SGX has no pigtail: **red = +**, **black = -**. F5 and F6 are both **MAXI 5006 50 A**. All battery / bus feeds are **10 AWG SGX** (labeled). 12 V feed from F6 is **red** to match DCIS output + through the mated DJ7021-8-21.

ID	Size / type	Color	End A	A termination	End B	B termination
12-BAT+	10 AWG SGX	red	BT1 + (12 V vehicle battery)	battery + post	F5 (MAXI 50 A) <i>and</i> F6 (MAXI 50 A) (splice)	MAXI holder lug
12-HOT	10 AWG SGX	red	F5	MAXI holder lug	Blue Sea Always_Hot bus <i>and</i> K3 (Cole Hersee 24213) pin 4	#10-32 stud / solenoid stud
12-SW	10 AWG SGX	red	K3 pin 3	solenoid stud	Blue Sea Switched bus	#10-32 stud
12-DJ+	10 AWG TXL	red	F6	MAXI holder lug	DJ7021-8-1 left top	8.0 mm contact
12-BAT-	10 AWG SGX	black	BT1 -	battery - post	Blue Sea Battery- <i>and</i> DJ7021-8-1 left bottom <i>and</i> chassis GND	#10-32 stud / 8.0 mm / chassis

DJ7021-8-1 left mates to right: 12 V battery +/- ↔ DCIS 12 V output +/- (red / black).

Ignition / inertia (coil side of K3). No factory harness; **red** = switched 12 V, **black** = ground.

ID	Size / type	Color	End A	A termination	End B	B termination
12-IGN1	18 AWG TXL (unlabeled)	red	SW3 (Ignition) A	switch terminal	K3 pin 2	Cole Hersee coil QC
12-IGN2	18 AWG TXL (unlabeled)	red	SW3 B	switch terminal	SW4 (Inertia Switch) A	switch terminal
12-KEY	18 AWG TXL	red	SW4 B	switch terminal	Blue Sea Key_Switch circuit	#8-32 screw, ring/spade
12-K3GND	18 AWG TXL (unlabeled)	black	K3 pin 1	coil QC	chassis GND	ring to chassis

K4 is **Omron G9EJ-1-E-UVD DC12** (schematic value). Coil is 12 V continuous on the READY rail (TXL), 100 mA / 1.2 W. Contacts close pack + onto the DCIS key (HV, signal current). Polarized contacts: pack + on terminal **3 (+)**, key on terminal **4 (-)**. Coil has no polarity. 0.250" QC.

ID	Size / type	Color	End A	A termination	End B	B termination
12-K4C	18 AWG TXL (unlabeled)	black	K4 pin 1	G9EJ coil QC	chassis GND	ring to chassis
12-K4S	18 AWG CWHMIO	blue (READY, pin 2)	K4 pin 2	G9EJ coil QC	READY net (WP-R3)	see §7

6. Charge-control 12 V (RAC02 + T92 coil)

RAC02 has no colored pigtail. **Orion CWHMIO 18 AWG colors carry** onto the T92 coil.

ID	Size / type	Color	End A	A termination	End B	B termination
CHG-12+	18 AWG CWHMIO	red (CHARGE, pin 3)	RAC02 +VOUT	RAC02 DC pin	T92 A1 and Orion CHARGE_Power pin 3	0.250" QC / TE 1376360-1 pin 3
CHG-12-	18 AWG (unlabeled)	black	RAC02 -VOUT	RAC02 DC pin	chassis GND	ring to chassis
CHG-SAFE	18 AWG CWHMIO	red/white (Charger_Safety, pin 6)	T92 A2	0.250" QC	Orion Charger_Safety_(on/off) pin 6	TE 1376360-1 pin 6

7. Delphi Weather Pack 4 (12 V / BMS bulkhead)

4-way Weather Pack between the Blue Sea / K2 side and the Orion / CAN-isolator side. **Orion 18 AWG CWHMIO colors carry through both housings**. Schematic pin order is top → bottom.

Right housing (BMS / isolator) — Orion pigtail side

ID	Size / type	Color	Weather Pack pin	End B	B termination
WP-R1	18 AWG CWHMIO	black (Ground, pin 12; verify)	right pin 1 (top)	BB CANOP GND and Orion Ground pin 12	isolator / TE 1376360-1 pin 12
WP-R2	18 AWG CWHMIO	green (Always_On, pin 1)	right pin 2	Orion Always_On_Power pin 1	TE 1376360-1 pin 1
WP-R3	18 AWG CWHMIO	blue (READY, pin 2)	right pin 3	Orion READY_Power pin 2 and BB CANOP 12V+ and K4 pin 2	TE 1376360-1 pin 2 / isolator / G9EJ coil
WP-R4	18 AWG CWHMIO	blue/white (DISCHARGE_enable, pin 7)	right pin 4 (bottom)	Orion DISCHARGE_enable pin 7	TE 1376360-1 pin 7

Left housing (fuse block / K2) — same colors through the mate

ID	Size / type	Color	End A	A termination	Weather Pack pin
WP-L1	18 AWG	black (from pin 12)	Blue Sea GND circuit	#8-32 screw	left pin 1 (top)
WP-L2	18 AWG	green (Always_On)	Blue Sea Always_On circuit	#8-32 screw	left pin 2
WP-L3	18 AWG	blue (READY)	Blue Sea Ready/Precharge and K2 pin 1	#8-32 / T9A QC	left pin 3
WP-L4	18 AWG	blue/white (DISCHARGE_enable)	K2 pin 2	T9A QC	left pin 4 (bottom)

Left mates to right, pin-for-pin.

8. CAN

Schematic: CAN1 250 kbps; **CAN shielding not terminated** on the inverter Deutsch. Orion CAN1 is a **12 ft shielded twisted pair** on the CWHMIO. Orion shield (pin 17) lands on the isolator. Shield grounded at one end only (Orion wiring manual).

Trunk color = Orion CAN1 pair. TSM2500 **green/white** (H) and **blue/white** (L) and CANdapter pigtails splice onto that pair and keep factory color only up to the splice.

8.1 HyPer 9 to Deutsch DTM 3

HyPer 9 K1 is AMPSEAL 776164-1. CAN_H = K1-13, CAN_L = K1-2. Standard K1 kits often leave K1-13 unpopulated — identify by **pin number**; colors vary.

ID	Size / type	Color	End A	A termination	End B	B termination
CAN-INV-H	18–20 AWG K1	K1-13 harness — verify	HyPer 9 CAN_H (K1-13)	AMPSEAL pin 13	Deutsch DTM left pin 2	DTM socket
CAN-INV-L	18–20 AWG K1	K1-2 harness — verify	HyPer 9 CAN_L (K1-2)	AMPSEAL pin 2	Deutsch DTM left pin 3	DTM socket
CAN-INV-SH	not terminated	—	Deutsch DTM left pin 1	DTM socket	—	schematic: shield not terminated

HyPer 9 **CAN_H_RES** (K1-14) is jumpered to **CAN_L_RES** (K1-3) on the schematic with note **internally termination**. That is the inverter termination jumper, not a harness cable.

8.2 Deutsch DTM 3 to BB CANOP (inverter side of isolator)

Carry the same K1 CAN colors through the DTM.

ID	Size / type	Color	End A	A termination	End B	B termination
CAN-ISO-1	same as CAN-INV-*	carry K1 CAN colors	Deutsch DTM right pin 1	DTM pin	BB CANOP pin 1	isolator CAN TB
CAN-ISO-2	same as CAN-INV-H	carry K1-13	Deutsch DTM right pin 2	DTM pin	BB CANOP pin 2	isolator CAN TB
CAN-ISO-3	same as CAN-INV-L	carry K1-2	Deutsch DTM right pin 3	DTM pin	BB CANOP pin 3	isolator CAN TB

8.3 BB CANOP (BMS side) to CAN1 bus

ID	Size / type	Color	Ends	Terminations
CAN1-H	18 AWG CWHMIO CAN1 pair	Orion CAN1_H (pin 18; <i>verify inner</i>)	BB CANOP pin 6; Orion CAN1_H pin 18; CANdapter CAN_H ; TSM2500 CAN_H (green/white pigtail to splice only)	isolator; TE 1376360-1 pin 18; CANdapter DB9 pin 7; charger CAN
CAN1-L	18 AWG CWHMIO CAN1 pair	Orion CAN1_L (pin 19; <i>verify inner</i>)	BB CANOP pin 7; Orion CAN1_L pin 19; CANdapter CAN_L ; TSM2500 CAN_L (blue/white pigtail to splice only)	isolator; TE 1376360-1 pin 19; CANdapter DB9 pin 2; charger CAN
CAN1-SH	shield drain	drain / bare	BB CANOP pin 5	Orion CAN1_Sheild pin 17. Do not ground the shield elsewhere.

8.4 CANdapter USB

ID	Size / type	Color	End A	A termination	End B	B termination
USB-1	USB 2.0 cable	black jacket	CANdapter USB	USB-B on adapter	USBCConnector1	USB connector (vehicle / laptop)

USBCConnector1 has a second pin with a short unterminated stub on the schematic (no far-end device).

9. J1772 pilot / proximity (Orion)

CWHMIO **18 AWG** colors carry from the I/O connector to the inlet.

ID	Size / type	Color	End A	A termination	End B	B termination
J1772-CP	18 AWG CWHMIO	orange/white (Control_Pilot, pin 13)	Orion J1772_Control_Pilot pin 13	TE 1376360-1 pin 13	J1772 Control Pilot	J1772 CP pin
J1772-PP	18 AWG CWHMIO	brown/white (Proximity_Detect, pin 14)	Orion J1772_Proximity_Detect pin 14	TE 1376360-1 pin 14	J1772 Proximity Detect	J1772 PP pin

10. Contactor coils (HyPer 9 K1)

Use the **K1 harness color printed for that pin** (kits vary). No published color for K1-25 / K1-26 in the HyPer text manual.

ID	Size / type	Color	End A	A termination	End B	B termination	Source
MC-X1	18–20 AWG K1	K1-25 harness — verify	HyPer 9 COIL_RETURN_+ (K1-25)	AMPSEAL 776164-1 pin 25	G1 X1	Gigavac coil	HyPer K1
MC-X2	18–20 AWG K1	K1-26 harness — verify	HyPer 9 DRIVER_OUT_1_– (K1-26)	AMPSEAL 776164-1 pin 26	G1 X2	Gigavac coil	HyPer K1

Not wired on this sheet

Item	Notes
Orion Fan_Monitor_MPI3 pin 9, Fan_Enable_MPO3 pin 10	No-connect (leave CWHMIO tails capped)
Blue Sea 5032 fused circuits 1–4 and 6	Unused (circuit 5 is Key_Switch / 12-KEY)
Tesla module cell taps	Separate Orion tap harness (22 AWG, orange / red / yellow, black grounds)
HyPer 9 motor phase / encoder / thermistor	Encoder: red / yellow / green / black as in HyPer manual
TSM2500 12 V aux, LED, drive-away, temp	Not on this schematic

K4 is Omron **G9EJ-1-E-UVD DC12** (see §5).

CWHMIO pins still to read off the card

Pin	Function	Color (from card)
12	Ground (expect black)	
18 / 19	CAN1_H / CAN1_L inner	

Schematic check (2026-08-23)

Parsed **IH53EV.kicad_sch** after F5 / SGX / color updates. Gaps left as documented:

- Traction 2/0 is **orange** with red/black heatshrink (vehicle convention; schematic stroke may still show +/-).
- Vehicle DCIS pack HV is **red / black** from the 3-way; factory pigtail stays yellow / gray.
- F5 and F6 are both **MAXI 5006 50 A**; battery feeds are **10 AWG SGX** (sheet wire key still says “SGX = 4 AWG”).
- Precharge PC-01/02/03 has no size label; assigned **18 AWG HV** (T9A QC). Circuit_Breaker2 sheet text is **1Pole_1000V_DC_10Amp**.
- RAC02 / F7 taps: schematic has both **18AWG AC** and **18AWG 300VAC**; same runs as the wire key.
- DC-IN+ is one net with the K4 tap: **10 AWG HV** on breaker → stud, **18 AWG HV** at the tap (DC-K4TAP).
- Circuit_Breaker3 / Circuit_Breaker4 / 2-pole breakers have empty Value fields; ratings are from the reference / nearby text (32 A charger, 10 A DCIS).
- J1772 inlet instance reference is **J1773**.
- Unused: Blue Sea circuits 1–4 and 6, Orion fan pins 9/10, Tesla cell-tap pins, one TerminalBlock3 1/4” stud, USBConnector1 stub.